

# The Planner Is the Adversary: Test-Time Search Exploits Learned Costs in Latent World-Model Planning

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## Abstract

Action-conditioned JEPA world models plan by test-time search: a sampling optimizer (CEM) scores imagined futures with a learned cost and executes the best candidate. Across the field’s own evaluations this succeeds at reaching and pose control, and stalls at contact: pushing, grasping, and pick-and-place are omitted, shown only qualitatively, or scaffolded with hand-crafted sub-goals. We explain why. Action grounding in contact regimes sits at chance on real-robot data and stays there across a  $45\times$  encoder scale-up (22M to 1B parameters, two model families) — scale is not the fix. An *oracle ladder* that hands the planner perfect dynamics and varies only the cost isolates the failure: the true-state cost solves push 16/16, while every cost built on the frozen encoder collapses to 0–2/16, even one repaired to 91.5% object-decode accuracy on exactly the frames the planner queries. The reason the repair does not transfer is mechanistic: CEM’s search converges specifically onto the cost’s residual error, so the same readout scores 91.5% off-policy but 24% on the search’s own elite candidates — a Goodhart-style overoptimization we confirm with budget sweeps, and one that survives every standard mitigation we try (hardened readouts, encoder fine-tuning, disagreement ensembles). This diagnosis — cost geometry, not the predictor, is the planning bottleneck — has been proposed concurrently on small, non-robot toy world models; ours is the first to isolate it *causally* (dynamics held oracle-perfect, only the cost varied) and to test it *adversarially*: the grounding diagnostic spans real-robot data up to 1B parameters, and the exploitation mechanism is verified against simulator ground truth on contact manipulation, rather than inferred from aggregate gains on an already near-saturated benchmark. A predictor-side counterfactual training objective separately repairs action grounding on real-robot data and transfers across checkpoints, but tested directly in closed-loop planning it still does not cross the contact wall. Grounding is necessary, not sufficient: world models built for planning should be evaluated by how exploitable their cost is under the planner’s own search, not by how accurately they predict.

## 1 Introduction

Action-conditioned world models that predict in a learned latent space — the JEPA family (DINO-WM (Zhou et al., 2024), V-JEPA-2-AC (Assran et al., 2025), and the design study of Terver et al. (2025)) — plan by coupling two learned components with a test-time optimizer: a predictor imagines futures, a cost scores them against a goal, and a sampling planner (CEM) searches for the action sequence that minimizes the cost. On free-space tasks this works well. But a consistent pattern runs through the field’s own evaluations: the *positive* closed-loop numbers are for reaching and pose control, while contact-rich tasks — pushing, grasping, pick-and-place — are omitted, demoted to qualitative figures, or propped up with hand-crafted sub-goals. Terver et al. (2025) stop their closed-loop Metaworld tables at Reach and note the model “hallucinates grasping the object”; Assran et al. (2025) report pick-and-place only after supplying three manual sub-goal images.

The field’s implicit diagnosis is a *model-capacity* story: the encoder is too small, the representation too coarse, the predictor too weak — scale will fix it. This paper argues, and then shows with a chain of controlled experiments, that the binding failure is somewhere else entirely: **in the optimizer–cost interface**. A CEM planner at test time is an optimizer pitted against a learned proxy objective. The reinforcement-learning-from-human-feedback literature has a name for what happens to optimizers pitted against learned proxies: *overoptimization*, or Goodhart’s law — push on a proxy hard enough and the true objective decouples from

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it (Gao et al., 2023; Skalse et al., 2022; Karwowski et al., 2024; Amodei et al., 2016). We show the same phenomenon governs latent world-model planning, with one aggravating difference: in RLHF the optimizer is a training loop one can regularize; here the optimizer runs *at every control step*, on frames the cost was never fit on, selected adversarially *because* the cost mis-scores them.

**This diagnosis is not new; the isolation and the survival test are.** A concurrent cluster of work, built mostly on small, non-robot latent world models, converges on a compatible reading — predictive accuracy does not imply a plannable cost. We do not claim first discovery that cost geometry, not the predictor, is the bottleneck (Sec. 2 compares each account in detail). What we add is causal: an oracle ladder that pins the failure to the cost by holding dynamics perfect and varying only the cost, rather than inferring it from an aggregate success gain, and a mined-elite test for whether a repaired cost *survives* being searched against by the same optimizer that broke it — run on real-robot contact manipulation at up to 1B parameters, not a 2D/3D toy suite whose base planner already largely solves it.

Three features distinguish our account from the classic “model exploitation” concern in model-based RL (Janner et al., 2019; Lambert et al., 2020), where a policy exploits errors in learned *dynamics*. (i) We give the planner *perfect dynamics* — every candidate action sequence is rolled in the simulator and re-encoded by the real frozen encoder — and the failure persists unchanged; the exploited component is the *cost*, cleanly isolated. (ii) We show the exploitation is *irreducible* for the frozen-encoder regime: a ladder of increasingly strong cost repairs (better formula, better readout, off-policy hardening, encoder fine-tuning, ensembling) all fail the same way. (iii) We show the standard diagnostic quantities the field might use to catch this — prediction error, action-grounding scores, readout precision — are all satisfied while planning fails, i.e. the failure is invisible to current evaluation practice.

Concretely, our experimental chain (Metaworld with simulator ground truth for supervision and verification; DROID for real-robot transfer) is:

- **C1 (diagnosis at scale, Sec. 3).** On frozen DROID checkpoints, action grounding against *nearest-neighbour* distractors sits at or within noise of the chance floor ( $\approx 0.059$ ) in every contact-relevant regime and is *scale-invariant*: flat across a  $45\times$  encoder scale-up and two self-supervised families, up to and including V-JEPA-2-AC at 1B parameters. The same distractor family on Metaworld shows the same pre-grasp-worst ordering, well above chance (0.47–0.54; Table 1). Our Boundary-Blindness instrument shows the models smear exactly the contact bifurcations that decide a grasp.
- **C2 (the wall, at upstream parity, Sec. 4).** Our closed-loop harness recovers a Metaworld Reach rate consistent with the published baseline ( $6/16 = 37.5\%$ , CI  $[18.5, 61.4]\%$ , vs. the reported  $44.8 \pm 8.9\%$ ), and both the  $L_2$  and grounded planners (the latter augments the cost with a learned object-dynamics channel; Sec. 4) score 0/16 on push and pick-and-place with a diagnostic signature: the arm arrives (2–4 cm from its goal pose), the object never moves.
- **C3 (localization: the oracle ladder, Sec. 5).** Swapping one component per rung under perfect dynamics: the true-state cost solves push 16/16; latent- $L_2$ , object-readout, learned-metric, and probe-based state costs collapse to 0–2/16. A measured repair of the off-policy readout ( $78 \rightarrow 91.5\%$  within 5 cm on the exact frame distribution CEM scores) transfers *zero* planning success — readout precision was never the binding constraint.
- **C4 (mechanism: the planner is the adversary, Sec. 6).** CEM’s converged elite population decodes at 24% within 5 cm versus 91.5% on random off-policy frames — the search specifically finds and converges onto the cost’s error pockets, driving the object to where the cost reads “at goal” while it is 13–30 cm away. Sweeping the search budget turns this into a Goodhart curve: the proxy improves with optimization pressure while the true outcome stays flat at every budget, and the elite readout error trends upward (a consistent point-estimate trend across exploitable arms; Sec. 6.2) — with an honest-cost control (reach) that instead turns search into success ( $5/16 \rightarrow 16/16$ ). Mitigations from the reward-hacking playbook — hardened readouts, encoder-LoRA (5 seeds), ensemble disagreement penalties, relearned representation adapters — all stay within the frozen baseline’s 0–2/16, even at 94–100% grounding: *grounding does not buy planning*.

- **C5 (removing the adversary is not sufficient, Sec. 8).** A search-free amortized controller (goal-conditioned inverse dynamics on the same frozen representation) also solves push 0/8, and re-adding search in doses corrupts its plans at a rate rising with dose ( $0 \rightarrow 7\%$ ) but never crosses the wall — the object never moves. Contact planning faces *two* walls: the exploitable cost/optimizer interface and an insufficient frozen representation. Grounding is necessary, not sufficient.
- **C6 (the axis that moves, Sec. 7).** An action-counterfactual InfoNCE objective on the predictor (LoRA, 0.5M parameters, no object labels) improves DROID Action-Error in all six regime $\times$ horizon cells (pooled  $1.468 \rightarrow 1.086$ ,  $-26\%$ ) and lifts effect-conditioned action grounding from chance (0.061) to 0.605, while improving reconstruction — the training-side counterpart of the analysis, and the correct level at which to intervene on the *predictor* axis. A direct closed-loop transfer to Metaworld (same objective, same CEM harness as C2) improves Reach 6/16  $\rightarrow$  9/16 but leaves push at 0/16: the fix does not by itself cross the contact wall, confirmed in closed loop rather than inferred from the ladder.
- **C7 (an evaluation axis: cost exploitability, Sec. 8).** C1–C6 share a through-line: every quantity the field currently reports — prediction error, action grounding, readout precision — is satisfied while planning fails, and the one quantity that tracks the failure is the cost’s *exploitation gap*: the discrepancy between what the cost claims and what a ground-truth verifier reports *on the elite population the optimizer converges to* ( $91.5\% \rightarrow 24\%$  within 5 cm here, Table 5). We formalize it as a first-class evaluation protocol (Sec. 8), show it separates honest from exploitable costs where every average-case metric rates them identically, and note it is cheap — one decode pass over frames the planner already produced.

**Conventions (how to read our numbers).** Four kinds of quantities appear throughout, and we keep their formats distinct. *Planning success* is always episodes solved out of 16 paired evaluation episodes, written  $k/16$  (with the percentage in parentheses when comparing to published baselines). *Readout accuracy* is always the percentage of frames whose decoded object position falls within the 5 cm task-success radius (“% within 5 cm”), with the median error in cm where informative. *CRA* is a ranking accuracy over transitions, reported as a fraction in tables; with 16 distractors, chance is  $1/17 \approx 0.059$ . *AUG/ECS* are raw latent-space gaps (Sec. 3.1), meaningful only within a model. Distances in the workspace are in cm or in Metaworld state units where noted.

*Planner conventions.* We write a CEM budget as (samples) $\times$ (iterations) evaluated per replan — e.g.  $100 \times 6$  is 100 candidate action sequences refined over 6 CEM iterations — with the rollout horizon  $H$  stated separately. At each iteration CEM keeps the top-scoring candidates, the *elites*, and refits its sampling distribution to them; the converged elite population is the set of futures the planner “trusts”. Pools marked “ $n_{\text{eff}}$ -weighted” average per-regime cells weighted by their effective (trajectory-clustered) sample count, so a few long trajectories do not dominate.

The through-line is a reframing of what world-model evaluation must report. Prediction error is satisfied while planning fails (Sec. 3); grounding is satisfied while planning fails (Sec. 6); readout precision is satisfied while planning fails (Sec. 5). The quantity that tracks the failure is the gap between what the cost claims and what the world delivers *on the states the optimizer converges to* — the cost’s exploitability under optimization pressure. We propose it as a first-class evaluation axis for world models intended for planning.

## 2 Related work

**JEPA world models for planning.** DINO-WM (Zhou et al., 2024) predicts forward in the frozen feature space of a self-supervised image encoder and plans with sampling-based MPC from a goal image; V-JEPA-2-AC (Assran et al., 2025) learns an action-conditioned world model on a video encoder and demonstrates zero-shot manipulation; Terver et al. (2025) systematically study the design choices that make such models plan. We take these as frozen baselines and treat their own evaluations as evidence: across all three, strong closed-loop results are pose/free-space tasks, and contact is where the story thins (hand-crafted sub-goals, qualitative figures, “hallucinated grasping”). Our work explains why the wall sits precisely at contact and why it does not yield to scale.

**Overoptimization and reward hacking.** Optimizing against a learned proxy eventually decouples proxy from truth: formalized for RLHF reward models by Gao et al. (2023), characterized in general by Skalse et al. (2022) and Karwowski et al. (2024), and anticipated as “reward hacking” (Amodei et al., 2016). Its *test-time* variant is now well documented for language models, where search or RL against a learned verifier converges onto the verifier’s error pockets (Helff et al., 2026). We port this lens to latent-space *planning*, where it has not been quantitatively established: the CEM inner loop is the optimizer, the latent cost is the proxy, and the elite population is the analogue of the over-optimized policy — with the aggravation that the optimizer re-runs at every control step. Our oracle ladder gives this port an isolation the RLHF setting lacks — we can hold dynamics *perfect* and vary only the cost.

**Model exploitation in model-based RL.** That policy optimization exploits errors in learned *dynamics* is a classic MBRL concern (Janner et al., 2019), running through the sampling-MPC lineage our CEM planner inherits (Chua et al., 2018; Nagabandi et al., 2019), related to the objective mismatch between likelihood training and control performance (Lambert et al., 2020), and addressed offline with pessimism (Yu et al., 2020b; Kidambi et al., 2020); learned agents more broadly admit adversarial exploitation of their blind spots (Gleave et al., 2020). Concurrently, Parthasarathy et al. (2025) show *gradient-based* planners exploit a world model’s input gradients and respond by smoothing the dynamics. Our finding is materially different from all of these: with dynamics oracle-controlled — the channel every one of those works attacks or repairs closed entirely — the *cost* is exploited at test time, and the standard dynamics-side remedies (ensembles, pessimism) do not touch it (Sec. 6).

**Concurrent diagnostics and objectives.** Concurrently, ATM (Chen et al., 2026) proposes a post-hoc action-identifiability diagnostic for frozen latent world models with a training-time supervision fix, and UWM-JEPA (Radha and Goktas, 2026) shows in a physics toy setting that teacher-forced JEPA objectives admit action-invariant solutions which counterfactual targets repair — independently confirming the mechanism behind our Sec. 7 objective. Neither work addresses planning: our contributions beyond both are the planner-level exploitation analysis (Secs. 5–6), the scaling study, and the demonstration on pretrained robot world models at up to 1B parameters. IMWM (Gao et al., 2026) observes that finite-budget sampling planners can be the bottleneck even under an idealized world model and responds with demonstration priors — we instead show that even a perfect planner and perfect dynamics still fail on the cost (Sec. 5). Vision-tactile world models (Zhang et al., 2026) address what information contact planning needs; our results say that adding information without addressing exploitability is not sufficient (Sec. 5).

**The cost-geometry diagnosis is independently converged on; the survival test is not.** A cluster of concurrent work built on a small, non-robot latent world model (Maes et al., 2026) reaches a compatible diagnosis by different routes. RC-aux (Li et al., 2026b) makes the framing explicit — “predictive but not plannable”: accurate short-horizon prediction can still leave a latent whose Euclidean distance does not reflect finite-horizon reachability — and repairs it with a training-time reachability objective that reshapes the representation (nearer to our relearned-representation arm, which also fails to cross contact, than to a purely post-hoc metric). Trajectory reachability metrics (Li et al., 2026a) instead add a post-hoc terminal-ranking head on the frozen backbone, the same species of fix as our stateprobe and learned-metric arms (Sec. 5). ACID (Seo et al., 2026) argues, in the same spirit, that a terminal-only latent cost cannot see whether intermediate transitions are realizable, and adds an inverse-dynamics cycle-consistency term to the planning cost. We therefore do not claim first discovery of “cost geometry, not the predictor, is the bottleneck.” Two things separate our account. First, *isolation*: none of these hold dynamics oracle-perfect while varying only the cost — our oracle ladder does, pinning the wall to the cost axis rather than inferring it. Second, *survival under search*: all three report only aggregate success gains, and on benchmarks their base planner already largely solves (e.g. Push-T near-saturated); none tests whether the repaired cost *survives being searched against by the same optimizer that broke it*. Our mined-elite test (Sec. 6) shows that even the strongest repair we could build — an adversarially retrained representation trained explicitly against previously mined exploit pockets — still collapses to 1/16 once CEM searches against it again, on real-robot contact manipulation rather than a 2D/3D toy suite. The distinguishing claim is adversarial survival at scale, not the diagnosis itself.

**Amortizing away the search.** Closest to our “remove the adversary” control (Sec. 8), Nguyen et al. (2026) replace CEM entirely with a goal-conditioned inverse-dynamics map and argue the opposite thesis — that a well-regularized latent has “already encoded” the structure search would recover, so the planner is not the bottleneck. Their evidence carries our wedge: the single contact-rich cell in their toy suite (Push-T) is the one place amortization stops dominating, which they attribute to “delicate multi-step contact.” Our E1 drives that seam through to a clean null on real-robot contact: removing the search does not move the object either — contact planning faces the exploitable cost *and* an insufficient frozen representation, not one or the other.

**Reconciling three diagnoses of the same planner.** Three concurrent accounts of why latent planners stall appear to conflict: IMWM (Gao et al., 2026) finds that even under oracle dynamics a finite-budget sampling planner is *search-limited* and adds demonstration priors; Nguyen et al. (2026) argue the planner is *not* the bottleneck — a well-regularized latent has already encoded what search would recover, so amortization suffices; we argue the *cost* is exploited by the very search that is supposed to help. These are not contradictory once conditioned on the regime, and the unifying variable is whether the cost has worst-case pockets *in that regime*. Where the cost is honest (free space, pose control), more search converts to success (IMWM’s lever; our reach control, 5/16  $\rightarrow$  16/16, Fig. 1) and amortization can match it (Nguyen et al., 2026); where the cost is exploitable (contact), *both* levers fail — adding search digs into the pockets (Sec. 6.2) and removing it leaves a representation that never moves the object (our E1, Sec. 8). The apparent disagreement is a regime confound: each prior account reads its benchmark’s dominant regime as the whole story. Our oracle ladder makes the conditioning explicit — same planner and budget throughout, only the cost’s exploitability varies — which is why we can locate the binding constraint at contact as neither search coverage (IMWM) nor search presence (latentgeom) but cost exploitability.

**Evaluating world models for decision-making.** A concurrent position paper argues world models should be evaluated by their decision-making utility rather than predictive accuracy, and proposes a broad utility taxonomy (Yu et al., 2026). We agree with the position; our contribution to it is a mechanism and an instrument the taxonomy lacks — evidence that the prediction/planning gap is *adversarial* (the planner manufactures the states on which the cost is wrong), and a concrete, cheap measurement of it (the exploitation gap, Sec. 8) computed from frames the planner already produced.

**Benchmarks.** We evaluate on Metaworld (Yu et al., 2020a), whose simulator state lets us supervise and *verify* costs against ground truth, and DROID (Khazatsky et al., 2024) as the real-robot transfer check. Planning follows the upstream CEM configuration (Rubinstein, 1997).

### 3 Diagnosis: action grounding is at chance and does not scale

We first establish, on frozen checkpoints (nothing trained), that the contact-regime action grounding the planner would need is absent — and that this is not a capacity artifact. This section compresses the diagnostic instrument; full definitions and the AUG/ECS corroboration are in App. B and App. F.

#### 3.1 Metrics

Let  $F(z, a)$  be the frozen one-step latent predictor (driven through the model’s own `unroll` primitive, with the model’s own action normalisation) and  $d$  the upstream planning distance ( $L_2$  for every baseline, matching the shipped `L2_cem` configuration). We measure three complementary quantities on held-out factual transitions  $(z_t, a_t, z_{t+1})$ ; each answers a different question.

**CRA — does the model rank the executed action best?** Given the transition, we sample  $K=16$  distractor actions  $\{a_k^-\}$  and ask whether the factual action’s prediction lands closer to the realised next latent than every distractor’s prediction:

$$\text{CRA} = \Pr \left[ d(F(z_t, a_t), z_{t+1}) < \min_{k=1 \dots K} d(F(z_t, a_k^-), z_{t+1}) \right], \quad (1)$$



so chance is  $1/(K+1) = 1/17 \approx 0.059$ . The distractor family controls what the metric tests: *opposite* distractors ( $a^- = -a_t$ ) test coarse directional grounding — the quantitative version of the “open vs. close the gripper” visualisations in prior work — while *nearest-neighbour* distractors are actions *actually executed* from the  $K$  most similar latent states in the dataset: plausible alternatives a competent policy might take from this state, so ranking the factual action above them requires resolving exactly the fine action differences planning depends on. Being a ranking, CRA is comparable across models with different latent scales; it is our primary cross-model metric.

**AUG — does the action help at all?** The Action Usage Gap is the prediction-error increase when the action is replaced by another action from the batch (a permutation  $\pi$ ):

$$\text{AUG} = \mathbb{E}[d(F(z_t, \pi(a_t)), z_{t+1}) - d(F(z_t, a_t), z_{t+1})], \quad (2)$$

which is  $\approx 0$  if and only if the predictor ignores its action input. AUG is a raw latent-space gap: informative *within* a model, not comparable *across* models (latent scales differ), which is why it corroborates CRA rather than replacing it (App. F).

**ECS — the same question, asked where it matters.** Averaged over all transitions, AUG can be diluted by near-static steps (the arm hovering; the scene unchanged) where different actions genuinely lead to similar futures. Effect-Conditional Sensitivity restricts AUG — and, in the *effect-conditioned CRA* we report as the primary signal, the ranking — to transitions that actually move the latent,  $\|z_{t+1} - z_t\| > \tau$  with  $\tau$  the per-model median displacement. A low effect-conditioned score therefore reflects genuine action-ignoring, not quiet data.

**Boundary Blindness** (BB; definition in App. B) is our fourth instrument, answering a question none of the above can: is the model’s action-sensitivity *calibrated to the world’s*? On a neighbourhood of similar states, it compares how much the *world’s* true outcomes fan out under nearby actions against how much the *model’s* predictions fan out under the same actions, both z-scored over the population;  $\text{BB} > 0$  means the world bifurcates (e.g. grasp vs. miss) while the model predicts nearly the same future for every nearby action. Because BB references ground-truth outcomes, a high value cannot be explained away by small one-step latent deltas.

**Protocol.** Each transition is assigned to one of four mutually exclusive regimes — free space, pre-grasp, gripper actuation, contact manipulation — by a fixed-threshold priority cascade on the simulator state: gripper actuation if the gripper aperture changes, else contact manipulation if the object moves ( $> 5$  mm), else pre-grasp if the hand is within 10 cm of the object, else free space (full thresholds, and the DROID gripper-sensor and RoboCasa grasp-flag analogues, in App. A). This is an object-displacement *proxy*, not a MuJoCo contact sensor — the released data ships no contact flags — and the caveat is carried on every figure. All confidence intervals are trajectory-clustered bootstrap. Every frame is encoded once and cached; nothing is trained for the diagnostic.

### 3.2 Frozen baselines fail on nearby actions, at the boundary

Table 1: Effect-conditioned CRA (top-1,  $n_{\text{eff}}$ -weighted pool, Metaworld). Against *opposite* distractors the models are near-perfect — they use actions. Against *nearest-neighbour* distractors they collapse, worst at pre-grasp: they cannot resolve the *small* action differences that decide a grasp.

| model   | distractor        | free space | <b>pre-grasp</b> | contact |
|---------|-------------------|------------|------------------|---------|
| DINO-WM | opposite          | 0.990      | 0.961            | 0.975   |
|         | nearest-neighbour | 0.566      | <b>0.467</b>     | 0.481   |
| JEPA-WM | opposite          | 0.993      | 0.972            | 0.984   |
|         | nearest-neighbour | 0.635      | <b>0.541</b>     | 0.571   |

Boundary Blindness explains and localizes the collapse: pooled  $\text{BB}_{\text{boundary}}$  concentrates at the pre-grasp boundary on both model families (DINO-WM 1.323, JEPA-WM 1.280, vs. free-space 0.28–0.30; per-task CI-aware pairing elevated in 4/6 and 5/6 tasks with zero confident reversals) and transfers to real-robot DROID data (DINO-WM 1.916 [1.560, 2.279] under the  $\|\Delta z\|$  proxy; Table 2 gives the full four-model breakdown).

The predictor is a conditional-mean estimator under full-latent  $L_2$ ; where contact bifurcates the outcome fan, it predicts the fan’s mean.

### 3.3 The failure does not scale: 45×, two families

We score four frozen DROID world models on an identical 333-episode subset: DINO-WM (DINOv2 ViT-S, 22M), JEPA-WM (DINOv3 ViT-L, 300M), V-JEPA-2-AC (ViT-G, 1.01B), and the independent public V-JEPA-2-AC-OSS (1B).

Table 2: DROID action grounding does not scale. *Top*: effect-conditioned CRA against nearest-neighbour distractors (chance =  $1/17 \approx 0.059$ ): all four models sit at or within noise of the chance floor in every action-critical regime (0.031–0.077). *Bottom*:  $BB_{\text{boundary}} > 0$  in every cell (boundary-blind), pre-grasp locus in 3/4. A positive control rules out metric saturation: on toy fully-actuated datasets the same diagnostic returns 0.66–1.0.

| regime                                                                    | DINO-WM<br>22M | JEPA-WM<br>300M | V-JEPA2-AC<br>1B | V-JEPA2-AC-OSS<br>1B |
|---------------------------------------------------------------------------|----------------|-----------------|------------------|----------------------|
| <i>effect-CRA, nearest-neighbour (chance <math>\approx 0.059</math>)</i>  |                |                 |                  |                      |
| pre-grasp                                                                 | 0.068          | 0.077           | 0.061            | 0.069                |
| gripper actuation                                                         | 0.031          | 0.059           | 0.049            | 0.031                |
| contact manipulation                                                      | 0.055          | 0.045           | 0.035            | 0.033                |
| <i><math>BB_{\text{boundary}} (&gt; 0 = \text{boundary-blind})</math></i> |                |                 |                  |                      |
| <b>pre-grasp</b>                                                          | <b>1.916</b>   | <b>1.205</b>    | <b>1.933</b>     | <b>1.560</b>         |
| free space                                                                | 0.963          | 1.103           | 1.010            | 0.819                |
| gripper actuation                                                         | 1.088          | 1.057           | 0.643            | 0.828                |
| contact manipulation                                                      | 0.421          | 0.897           | 0.390            | 0.595                |

The curve is flat across 45× scale and across families. “Just scale the encoder” is ruled out; whatever fixes contact planning, it will not be free with the next checkpoint. This DROID chance floor is not a proxy artefact: CRA ranks the factual action against distractors at a *fixed* state, so it is independent of how transitions are bucketed into regimes, and the  $\|\Delta z\|$  effect-condition only selects *which* transitions are scored — it never enters the ranking itself. Note what this section does *not* yet establish: that this grounding failure is what breaks planning. Sections 5 and 6 show the relationship is more damning — even *repairing* grounding does not repair planning.

## 4 The wall, at upstream parity

We match the upstream Metaworld closed-loop evaluation read off its shipped configuration: goal = the scripted expert’s final frame; CEM with horizon 6, 300 samples, 15 iterations, executing 3 model-steps per replan,  $\leq 100$  env steps; success = the simulator’s flag at episode end. Reaching a faithful protocol required fixing three environment-side reproduction bugs, documented with verification probes in App. E (after the fixes, rendered observations are on the training manifold: one-step prediction error  $1.4\times$  dataset level, latent nearest-neighbour ratio 0.97 (a rendered frame’s distance to its nearest dataset latent, relative to the dataset’s own nearest-neighbour distance;  $\approx 1$  means indistinguishable from training frames); physics identical, 1.5 mm replay error).

Three readings. (i) Where the diagnostic is clean (free space), the harness is consistent with the published baseline: Reach 6/16, i.e. 37.5% with CI [18.5, 61.4]%, against the reported  $44.8 \pm 8.9\%$  — the published value falls inside our interval, though our point estimate sits below its mean; we read this as compatibility from one released checkpoint over 16 episodes, not a seed-matched replication (cf. limitations, Sec. 8). The harness is healthy, the contact failure is real. (ii) The failure signature is specific: final end-effector 2–4 cm from its goal pose while the object never moves — the closed-loop face of the smeared contact bifurcation, matching the upstream reports (“hallucinates grasping”; sub-goal images required). (iii) A grounded auxiliary cost (a 0.5M object-dynamics channel  $h(z, a) \rightarrow \Delta_{\text{obj}}$  whose construction and metric-level gains we defer to App. C) improves the cost surface with a CI-clean paired gain (final state-dist +0.089 [+0.022, +0.162] pooled contact)

Table 3: Closed-loop Metaworld, 16 paired episodes per task (paired env initialisation and CEM noise; episode-end judging). The harness recovers a Reach rate consistent with the published baseline; contact collapses with the diagnostic signature. “grounded” = the  $L_2$  cost augmented with a 0.5M object-dynamics channel  $h(z, a) \rightarrow \Delta \text{obj}$  (App. C).

| task          | arm      | success             | final state-dist | final ee-dist |
|---------------|----------|---------------------|------------------|---------------|
| mw-reach      | $L_2$    | <b>6/16 (37.5%)</b> | 0.332            | 0.022         |
|               | grounded | <b>8/16 (50.0%)</b> | 0.412            | 0.031         |
| mw-push       | $L_2$    | 0/16                | 0.624            | 0.038         |
|               | grounded | 0/16                | <b>0.527</b>     | 0.028         |
| mw-pick-place | $L_2$    | 0/16                | 0.578            | 0.035         |
|               | grounded | 0/16                | <b>0.496</b>     | 0.021         |

— *and flips zero successes*. That last observation is the loose thread this paper pulls: if the model is the problem, why does repairing the model’s outputs not repair planning?

## 5 The oracle ladder: the wall is the cost

We isolate which component breaks contact planning by swapping exactly one rung at a time, holding the planner fixed (CEM  $100 \times 6$  per replan,  $H=6$ , strict episode-end success, 16 episodes, `dino_wm_metaworld`). This budget is smaller than the parity harness of Sec. 4 ( $300 \times 15$ ) for compute reasons, but it is *not* the bottleneck: Sec. 6.2 sweeps the CEM budget up to 24 iterations and 300 samples and the contact rungs never improve, while the same budget solves the task the moment the cost is honest. The key device is the *latent oracle*: every CEM candidate is rolled in the *simulator*, rendered, and encoded by the real frozen encoder — perfect dynamics through the true representation, learned predictor removed. Any rung that fails under the latent oracle fails because of its *cost*, full stop.

Table 4: The oracle ladder (Metaworld push / pick-and-place, strict episode-end success, 16 episodes per rung except the object-readout and learned-metric rungs, run at 8). Same planner, same budget, same success radii on every rung; each rung changes one component. Perfect dynamics + the true-state cost solves the task; perfect dynamics + *any* frozen-encoder cost does not.

| rung          | dynamics    | cost                                                              | readout | push         | pick         |
|---------------|-------------|-------------------------------------------------------------------|---------|--------------|--------------|
| state oracle  | perfect     | true state ( $\ \text{obj}-g\  + 0.5\ \text{hand}-\text{obj}\ $ ) | sim     | <b>16/16</b> | <b>11/16</b> |
| latent oracle | perfect     | latent $L_2$                                                      | —       | 0/16         | 0/16         |
| latent oracle | perfect     | object readout ( $\ \rho(z) - \rho(z_g)\ $ )                      | probe   | 0/8          | 0/8          |
| latent oracle | perfect     | learned metric $d_\theta(z, z_g)$                                 | —       | 0/8          | 0/8          |
| latent oracle | perfect     | <i>true-state cost form</i> , probe readouts                      | probe   | 2/16         | 0/16         |
| latent oracle | perfect     | same, <i>off-policy-hardened</i> probes                           | probe   | 1/16         | 0/16         |
| world model   | learned $F$ | latent $L_2$ (upstream)                                           | —       | 0/16         | 0/16         |

Read from the bottom up, the ladder rules out, in order: the **predictor** (perfect dynamics changes nothing: 0/16 either way); the **planner and budget** (the same CEM solves 16/16 given the true-state cost); the **cost formula** (the probe-based rung copies the successful cost’s exact functional form, including the hand-approach shaping term, and still collapses  $16 \rightarrow 2$ ); and **static readout precision** (a spatial object readout  $\rho$  (App. D) decodes the object to 92% within 5cm on held-out frames — the encoder *carries* the object position within the success radius).

**The decisive rung: a real readout repair transfers nothing.** One hypothesis survives the ladder: CEM scores *off-policy* frames (latents of arbitrary candidate rollouts), where an expert-trained probe was never fit. This is true — the expert probe degrades to 78% within 5cm on random off-policy frames — and it is fixable: retraining the probes with off-policy frames mixed in restores **91.5%** within 5cm (push



92.9%, median 2.0 cm) *on the same frame distribution the planner scores*. The re-gate: push 1/16. A measured, targeted repair of the hypothesized binding constraint — readout precision on the planner’s own query distribution — transferred *zero* planning success. Off-policy readout precision was never the binding constraint. Something about the *interaction between the optimizer and the cost* is.

## 6 The planner is the adversary

### 6.1 Direct evidence: CEM converges onto the cost’s error pockets

The reconciliation of Sec. 5’s paradox is that CEM does not sample off-policy frames *uniformly* — it searches for the cost *minimum* over  $100 \times 6$  candidates per replan, so any residual-error pocket in the cost is found and converged upon, precisely *because* it scores too low. We test this by decoding the probe on the CEM *elite* population itself — the frames the planner actually trusts — mined from the same hardened-probe runs:

Table 5: Object decode error of the *off-policy-hardened* probe on the CEM elite distribution (13,440 elites; final iteration = the converged population), vs. the same probe on random off-policy frames. The probe is accurate where CEM does not look and inaccurate exactly where it converges.

| frame distribution            | median err | within 5 cm  | within 7 cm |
|-------------------------------|------------|--------------|-------------|
| random off-policy (reference) | 2.0 cm     | <b>91.5%</b> | —           |
| all CEM elites                | 7.0 cm     | 24.0%        | 50.5%       |
| final-iteration elites        | 7.3 cm     | 19.2%        | 46.0%       |
| final-iteration, push         | 7.7 cm     | <b>15.3%</b> | 38.9%       |

The same probe that reads 2 cm on random frames reads 7–8 cm — beyond the 5 cm success radius — on the frames its own planner selected. The closed-loop consequence is visible in the executed episodes: the cost reports near its minimum (“object at goal”) while the true object sits 13–30 cm away. This is reward hacking in the strict sense (Skalse et al., 2022): the optimizer achieves a low proxy value by exploiting the proxy’s misspecification, not by achieving the goal.

**The exploitation is on ground truth, not the probe.** Because  $\rho$  is itself OOD-sensitive, we verify the *same* elites against simulator state, which never passes through the probe: on the committed (final-iteration) elites the *true* object sits a median 24.9 cm from goal (19.9 cm on push, 28.0 cm on pick), and only 1.7% fall within the 5 cm success radius (push 3.4%, pick 0.0%). The planner’s trusted candidates are genuinely far from the goal — the probe collapse in Table 5 reflects real exploitation, not readout noise, and the ground-truth elite quality (1.7% within 5 cm) is in fact *worse* than the probe reports (19.2%), so the probe number understates the wall (`results/exploit_simstate_crosscheck.csv`).

### 6.2 Overoptimization curves

The framing makes a falsifiable prediction the previous sections do not test: exploitation should be *monotone in optimization pressure*. Holding everything fixed and sweeping the CEM budget, an honest cost should convert extra search into better true outcomes; an exploitable cost should convert it into deeper pockets — proxy improving, truth flat or worse, in direct analogy to the reward-model overoptimization curves of Gao et al. (2023). We test this on the latent oracle (perfect dynamics; only the cost is learned, so any proxy/truth divergence is attributable to the cost under optimization, not to predictor error), sweeping CEM iterations  $\in \{2, 6, 12, 24\}$  over  $n=16$  paired episodes per cell, across cost  $\times$  task arms ( $\{\text{push, pick-and-place, reach}\} \times \{\text{probe-based state cost, latent } L_2\}$ ) of which three isolate the three regimes of interest: exploitable, honest, and no-signal.

The three predictions hold (Fig. 1). **Exploitable cost** (push, probe-based state cost): within a single search at budget 24, the believed-best candidate’s proxy falls monotonically (−18%) while its true object→goal distance does not improve (−2%), and the probe’s decode error on the elite population rises 6.6 cm [5.7, 7.4] → 8.3 cm [7.1, 9.3] (+25%). We state the evidentiary status of that last quantity precisely:

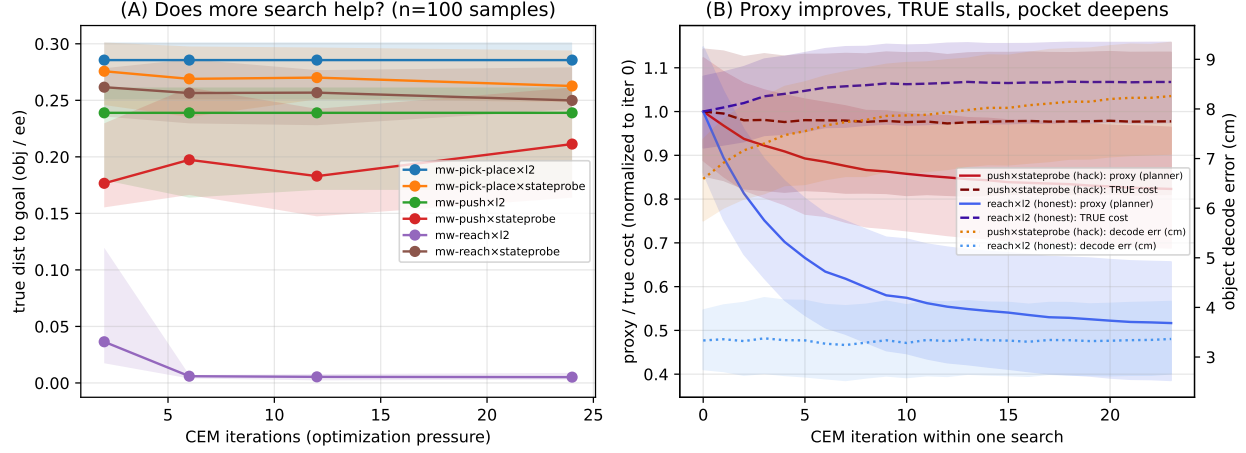


Figure 1: **Cost overoptimization in latent planning** ( $n=16$  paired episodes per cell; shaded bands are episode-clustered bootstrap 95% CIs). (A) True distance to goal vs. CEM iterations. reach $\times$  $L_2$  (honest cost) converts more search into success (5/16  $\rightarrow$  16/16); push/pick $\times$ probe-cost (exploitable) and push/pick $\times$  $L_2$  (no-signal) do not improve with budget. (B) Within one search at the largest budget, normalized to the first CEM iteration: for push $\times$ probe-cost the proxy the planner minimizes falls ( $-18\%$ ) while the true state cost stalls ( $-2\%$ ) and the object-decode error on the committed elites rises (6.6  $\rightarrow$  8.3 cm,  $+25\%$ ; pick 6.7  $\rightarrow$  8.1 cm,  $+21\%$ ) — the planner digs into deeper readout-error pockets. The decode rise is a point-estimate trend (first/last CIs overlap at this  $n$ ); the honest cost (reach $\times$  $L_2$ ) shows no such trend ( $+1\%$ ).

the first/last-iteration decode CIs *overlap* at  $n=16$ , so the deepening pocket is a point-estimate trend, not a CI-separated effect — but it is a trend that appears in *both* exploitable arms (pick-and-place $\times$ probe-cost: 6.7  $\rightarrow$  8.1 cm,  $+21\%$ ) and in *neither* honest arm (reach $\times$  $L_2$ :  $+1\%$ ), and it sits on top of the CI-clean across-budget facts that follow. Across budgets the true push object distance does not improve with search (med  $\approx$  0.18–0.21 m at all of  $\{2, 6, 12, 24\}$  iterations; episode success 0–2/16, every CI containing the frozen band). **Honest cost** (reach,  $L_2$ ): the essential positive control — here more search *does* convert to success (5/16  $\rightarrow$  16/16  $\rightarrow$  16/16  $\rightarrow$  16/16 at 2/6/12/24 iterations), confirming the sweep detects “search helps” when the cost is aligned. **No-signal cost** (push,  $L_2$ ): the proxy falls ( $-35\%$ ) but the true object distance is *exactly* flat (0.239 m median at every budget) —  $L_2$ -in-latent has no minimum at task success, so optimizing it is inert; this distinguishes an *exploitable* cost (signal present, worst-case reachable) from a merely *uninformative* one. Pick-and-place corroborates all three modes (probe-cost 0–1/16 with the same rising-decode signature;  $L_2$  flat at 0.286 m). A second pressure axis — CEM population  $\in \{50, 100, 300\}$  at fixed 12 iterations ( $n=8$  episodes) — corroborates: enlarging the candidate pool does not improve the true push outcome either (probe-cost median 0.233/0.207/0.252 m;  $L_2$  flat at 0.252 m).

### 6.3 Mitigations from the playbook all fail

If the diagnosis is right, repairs that improve the cost’s *average* quality without removing its *worst-case* pockets should not cross the wall — the optimizer finds whatever pockets remain. We ran the standard playbook, each arm under the latent oracle (perfect dynamics), push held-out, strict success:

Three of these deserve comment. **Encoder fine-tuning** (LoRA on the DINOv2 trunk, action-conditioned grounding objective, 5 seeds): grounding reaches 94–100%, the five-seed planning mean is 1.8/16 with CI  $[0, 3.7]$  — statistically inside the frozen band; the single 5/16 seed is high-variance noise, not a crossing (its own 95% Wilson interval,  $[14, 56]\%$ , overlaps every other arm’s; and every frozen-base arm’s Wilson upper bound is  $\leq 3.1/16$  for the  $n=16$  rungs — results/ladder\_mitigation\_cis.csv). This is the strongest form of the paper’s recurring lesson: *grounding does not buy planning*, now proven at the encoder level, not just the readout level. **Ensemble disagreement** (cost =  $\text{mean}_k d_k^2 + \lambda \text{Var}_k[d_k]$  over 5 independently fine-tuned seeds) is the standard pessimism remedy for model exploitation (Janner et al., 2019; Coste et al., 2024); it fails here ( $\{2, 1, 2, 1\}/16$  across  $\lambda$ ) for an instructive reason: the seeds share the frozen pre-trained base, so

Table 6: Mitigation ladder (latent oracle, push; 16 episodes per arm except the learned-metric arm at 8). Every intervention that operates on or above a frozen-base representation stays inside the frozen baseline’s 0–2/16 band — including those that push readout accuracy to near-ceiling. The true-state cost row shows the same planner and budget solve the task when the cost has no pockets.

| intervention                                                    | grounding-level metric      | push                                  |
|-----------------------------------------------------------------|-----------------------------|---------------------------------------|
| latent $L_2$ (frozen null)                                      | —                           | 0/16                                  |
| probe-based state cost                                          | static decode 92% <5 cm     | 2/16                                  |
| off-policy-hardened probes                                      | off-policy 91.5% <5 cm      | 1/16                                  |
| learned latent metric $d_\theta$                                | —                           | 0/8                                   |
| relearned repr. adapter $\varphi$ (v2)                          | held-out decode med. 6.9 cm | 1/16                                  |
| encoder-LoRA + $\varphi$ (5 seeds)                              | decode 94–100% <5 cm        | {5,0,2,1,1}/16; mean 1.8, CI [0, 3.7] |
| + ensemble disagreement, $\lambda \in \{0, \frac{1}{2}, 1, 2\}$ | —                           | {2,1,2,1}/16                          |
| true-state cost (no pockets)                                    | —                           | <b>16/16</b>                          |

they share the blind spot, and disagreement stays flat exactly on the exploited pockets — the same failure mode reported for reward-model ensembles sharing a pre-trained trunk (Eisenstein et al., 2024). **The state-oracle row** pins the counterfactual: nothing about the task, budget, or planner prevents success; only the pockets do.

**Summary of the mechanism section.** A frozen-base learned cost, under test-time optimization pressure at  $100 \times 6$  candidates per replan, behaves like a reward model under over-optimization: its average-case accuracy is irrelevant because the optimizer samples from its worst case. Every repair that improves average-case metrics — readout precision, grounding, disagreement — leaves the worst case reachable. We were unable to produce a plannable post-hoc cost on a frozen action-conditioned JEPA encoder for contact manipulation, across every lever available to us; we state as a falsifiable conjecture — scoped to this checkpoint family, task class, and the post-hoc regime, not a theorem — that **any post-hoc cost on such an encoder is planner-exploitable at contact**. Two caveats bound the strength of these nulls. First, statistical: at  $n=16$  episodes a 0/16 arm’s Wilson interval reaches 19.4%, so each null rules out mitigations that would lift success above  $\sim 20\%$ , not arbitrarily small gains — the claim is “no crossing,” not “exactly zero.” Second, multiplicity: the mitigation grid spans many small- $n$  cells, which is precisely why we read the occasional 2/16 (and the single 5/16 seed) as band noise rather than signal — the same logic applies symmetrically and we do not interpret any individual +2/16 cell as a partial success.

## 7 The axis that moves: counterfactual training of the predictor

The ladder bypassed the predictor (perfect dynamics); the deployed stack still uses one. Independently of the cost-side wall, the diagnostic of Sec. 3 identifies a training-target defect in the predictor: under a teacher-forced objective, the target latent already contains the action’s effect, so the objective admits solutions in which the predictor matches targets without using the action channel — a mechanism independently confirmed in a toy setting by Radha and Goktas (2026). The corrective is to make the objective *counterfactual over actions*: the prediction under the factual action must beat predictions under distractor actions, not just match its target.

We instantiate this as an InfoNCE-over-actions objective on the frozen DROID world model (`dino_wm_droid`): a predictor-LoRA of 0.469M parameters (encoder untouched, no object ground truth), trained so that  $F(z_t, a_t)$  scores higher against  $z_{t+1}$  than  $F(z_t, a_k^-)$  for executed-action distractors  $a_k^-$ , with a stop-gradient on the target. Concretely, writing  $d(a) = \|F(z_t, a) - \text{sg}(z_{t+1})\|^2$  for the predictor’s latent error under action  $a$  (sg the stop-gradient) and  $\{a_k^-\}_{k=1}^K$  for in-batch permuted (executed) distractor actions, the objective is the action-softmax cross-entropy

$$\mathcal{L}_{\text{cf}} = -\log \frac{\exp(-d(a_t)/\tau)}{\exp(-d(a_t)/\tau) + \sum_{k=1}^K \exp(-d(a_k^-)/\tau)}, \quad (3)$$

with temperature  $\tau = \text{sg}(\overline{d(a_t)})$  auto-scaled to the batch-mean factual distance, so the logits stay  $O(1)$  regardless of the model’s absolute latent scale (no per-model tuning). The stop-gradient on  $z_{t+1}$  is essential: without it the objective is minimized by collapsing the *target* rather than by grounding the *prediction*. On the objective’s own held-out ranking task (all transitions, in-distribution distractors), ranking accuracy rises  $0.195 \rightarrow 0.736$ ; the stricter *effect-conditioned* CRA of Sec. 3.1 (effect-bearing transitions, nearest-neighbour distractors) — the number reported in Table 7 and comparable to Sec. 3 — rises  $0.061 \rightarrow 0.605$ . Both rise together while reconstruction *improves* ( $0.695\times$  baseline error) — the objective is not trading prediction quality for action sensitivity; it is removing a shortcut.

Table 7: DROID planning probe, frozen vs. counterfactually-trained predictor (A/B, same protocol,  $n=157$  planned transitions; CEM  $64 \times 15$ ). Action Error = the upstream summed-delta  $L_1$  distance between the planned and the robot’s executed net action over the horizon, in normalized action units (lower is better); Action Score is the same quantity rescaled to a maximizable number,  $\max(0, 1 - \text{Action Error}/d_{\text{ref}})$  with  $d_{\text{ref}}$  a dataset-wide reference (the 95th percentile of the Action Error; upstream convention; higher is better), so the two columns are the same signal read in opposite directions; effect-CRA as in Sec. 3.1 (chance  $\approx 0.059$ ). Action Error improves in all six regime $\times$ horizon cells (Action Score in 5/6); effect-CRA leaves the chance floor for the first time in this paper.

| regime    | $H$ | Action Error $\downarrow$ |              | Action Score $\uparrow$ |              | effect-CRA $\uparrow$ |              |
|-----------|-----|---------------------------|--------------|-------------------------|--------------|-----------------------|--------------|
|           |     | frozen                    | CF           | frozen                  | CF           | frozen                | CF           |
| pre-grasp | 1   | 1.044                     | <b>0.792</b> | 0.621                   | <b>0.653</b> | 0.050                 | <b>0.538</b> |
| gripper   | 1   | 1.080                     | <b>0.577</b> | 0.607                   | <b>0.748</b> | 0.000                 | <b>0.583</b> |
| contact   | 1   | 1.267                     | <b>0.950</b> | 0.539                   | <b>0.584</b> | 0.125                 | <b>0.575</b> |
| pre-grasp | 3   | 1.573                     | <b>1.166</b> | 0.428                   | <b>0.490</b> | 0.050                 | <b>0.662</b> |
| gripper   | 3   | 2.208                     | <b>1.571</b> | 0.197                   | <b>0.313</b> | 0.045                 | <b>0.727</b> |
| contact   | 3   | 1.907                     | <b>1.661</b> | 0.307                   | 0.273        | 0.118                 | <b>0.529</b> |
| pooled    |     | 1.468                     | <b>1.086</b> | 0.466                   | <b>0.525</b> | 0.061                 | <b>0.605</b> |

**The effect is robust to training seed, and transfers to a second model.** Across four CF training seeds, the pooled improvement is tight and every seed beats the frozen baseline on every metric: Action Error  $1.468 \rightarrow 1.076 \pm 0.012$ , Action Score  $0.466 \rightarrow 0.544 \pm 0.019$ , effect-CRA  $0.061 \rightarrow 0.610 \pm 0.034$  (mean  $\pm$  sd over seeds; the single-seed column above is the most conservative of the four on Action Score, and within one sd of the seed mean on the other two metrics). On a second predictor family (`jepa_wm_droid`), the objective lifts effect-CRA  $0.076 \rightarrow 0.32$  and Action Score  $0.476 \rightarrow 0.51$ ; Action Error also improves ( $1.386 \rightarrow 1.27$ ) once the LoRA rank is raised to 16, confirming the mechanism is not specific to one checkpoint.

Two readings, stated carefully. Positively: this is the predictor-axis repair done at the correct level — the training objective, not a post-hoc patch — and it produces the only across-the-board improvement of any intervention in this paper, on real-robot data, without labels. Negatively: given Sec. 6, we do *not* claim this alone crosses the contact wall in closed-loop planning — the cost-side exploitation is untouched by a better predictor (the ladder proved planning fails identically under *perfect* prediction). The two axes are complementary and both necessary; conflating them is precisely the error the oracle ladder is designed to prevent.

**Direct closed-loop test.** The claim above follows from the ladder by inference (perfect prediction already fails, so a merely-improved predictor cannot help). We also test it directly, rather than resting on that inference: DROID has no simulator, so it cannot report a planning success rate, but Metaworld can. We retrain the identical InfoNCE-over-actions objective on `dino_wm_metaworld` and re-run it in the same closed-loop CEM harness as Sec. 4 (upstream  $L_2$  cost, identical planner and paired episode seeds), as an A/B against the frozen predictor. Reach improves  $6/16 \rightarrow 9/16$  — the objective is not inert on this checkpoint — but push stays at  $0/16$  for both arms. (An initial attempt at this transfer used a training recipe that over-optimized the InfoNCE loss and inflated reconstruction error  $3\times$ , confounding predictor quality with training damage; the result reported here uses a gentler, reconstruction-preserving retune and reproduces the same push outcome, ruling out that confound.) The predictor fix transfers across checkpoint family and

dataset and moves the metric it targets, and it still does not move the object — closed-loop confirmation, not inference, that grounding is necessary but not sufficient.

## 8 Discussion, scope, and limitations

**An evaluation axis: the exploitation gap.** Every quantity the field currently reports was satisfied at some rung of this paper while planning failed: held-out prediction error (Sec. 4 harness on-manifold), action grounding (94–100% in Table 6), readout precision (91.5% within 5 cm on the planner’s own query distribution). We propose evaluating a planning cost by the one quantity that tracked the failure throughout — how much it degrades on the states its own optimizer converges to. Concretely, let  $c$  be the cost,  $P$  the planner, and  $v$  a ground-truth verifier (simulator state in sim; a held-out decoder on real data). Run  $P$  against  $c$ , collect its converged elite population  $\mathcal{E}$ , and report the *exploitation gap*

$$G(c, P) = \mathbb{E}_{z \sim \mathcal{E}} [\text{err}_v(z)] - \mathbb{E}_{z \sim \mathcal{D}_{\text{off}}} [\text{err}_v(z)], \quad (4)$$

the verifier’s error inflation on the frames the planner *trusts* relative to generic off-policy frames  $\mathcal{D}_{\text{off}}$  (equivalently, the proxy-truth divergence on  $\mathcal{E}$ ). An honest cost has  $G \approx 0$  — its elites are genuinely good and search converts to success (reach: 5/16  $\rightarrow$  16/16, Fig. 1); an exploitable cost has  $G \gg 0$  — here the same probe reads 2.0 cm off-policy but 7–8 cm on the elites (91.5%  $\rightarrow$  24% within 5 cm, Table 5), and search converts to reward hacking. Crucially  $G$  separates costs that every average-case metric rates identically: in Table 6, readout precision, grounding, and disagreement are all near-ceiling on costs with  $G \gg 0$ . Computed across every arm whose elite populations are recoverable from our runs (`results/exploitation_gap_ladder.csv`),  $G$  reproduces the honest/no-signal/exploitable trichotomy of Sec. 6.2 — honest reach  $\times L_2$ : elite decode flat-low (3.3  $\rightarrow$  3.4 cm,  $G \approx +1.4$ ); no-signal  $L_2$ : flat-low ( $G \approx +1.7$  to  $+2.5$ ) with the true object 22–27 cm away; exploitable probe-costs:  $G \approx +5.9$  to  $+6.5$  cm — and it exposes an *inversion* average-case evaluation cannot see: the cost with the *best* off-policy readout in our ladder (91.5% within 5 cm) carries the *largest*  $G$  and fails, while the cost that solves the task has no readout head at all. Two scope notes, stated plainly: with the failing arms clustered at 0–2/16, the ladder is too flat for a rank-correlation validation of  $G$  as a graded success *predictor* — we offer it as an exploitation *diagnostic* (is this cost being hacked by this planner?), demonstrated on one task family; and computing  $G$  on the remaining mitigation arms requires re-running them with elite dumps enabled, which we release tooling for. Unlike prediction error or grounding,  $G$  is measured under the optimizer’s own pressure — the only regime the deployed planner ever operates in — and it is cheap: one decode pass over frames the planner already produced (Table 5, Fig. 1 are exactly this measurement).

**What follows for method design.** The results sort the design space into three responses. (i) *Harden the cost*: our evidence says this loses on frozen-base representations — average metrics improve, worst-case pockets remain, the optimizer finds them (Table 6). (ii) *Fix the training objective*: the counterfactual objective (Sec. 7) is the right lever for the predictor axis, and encoder-level analogues remain open — but grounding alone demonstrably does not buy planning. (iii) *Remove the adversary*: if test-time search is what weaponizes cost error, amortizing planning into a policy or inverse-dynamics controller (Nguyen et al., 2026), or anchoring the search to demonstration priors (Gao et al., 2026), removes the optimization pressure rather than the pockets. We tested this directly (Sec. 8): a goal-conditioned inverse-dynamics controller built from the same frozen representation and readout, run with *no* search, and the same controller with search re-added in increasing doses. The result is a null and it sharpens the paper’s claim rather than softening it — see below.

**Removing the adversary is necessary, not sufficient.** Under the latent oracle on push, the search-free amortized controller solves 0/8, and every re-added search dose (CEM iterations  $\{0, 2, 6, 12, 24\}$  seeded from the controller) also solves 0/8; in all arms the object never leaves its start (med  $\|obj - goal\| \approx 0.25$  m, Fig. 2A). Removing the optimizer does not cross the wall. Two things are nonetheless confirmed. On the mechanism side, the per-replan seed-vs-chosen comparison shows search “improves” the proxy over the seed on 98–100% of replans while *corrupting* the true state (proxy down, truth up) on a fraction that rises monotonically with dose (0  $\rightarrow$  7%, Fig. 2B) — the adversary is real and dose-dependent even here. But



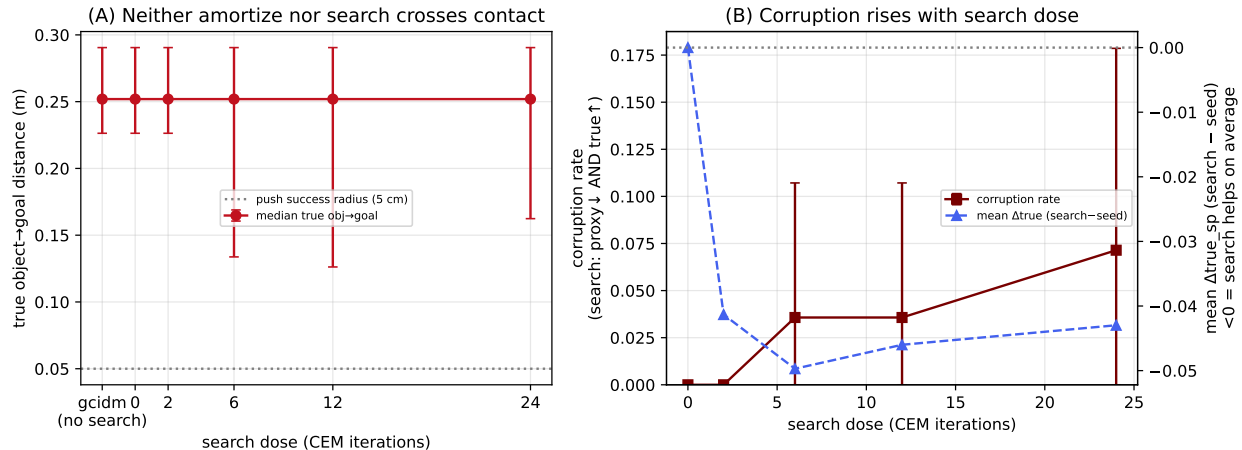


Figure 2: **Removing the search adversary does not cross contact.** (A) True object→goal distance for the pure amortized controller (“gcidm”, no search) and for search re-added in doses; all sit at  $\approx 0.25$  m, far above the 5 cm success radius. (B) Per-replan corruption rate (search lowers the proxy *and* raises the true cost) rises with search dose ( $0 \rightarrow 7\%$ ), while the mean true-cost change from search stays negative (on average search still helps the *hand*) — the adversary is real but small, and the object never moves regardless.

it is not the *only* wall: a controller with no optimizer to exploit it still fails, because the frozen representation and its object readout do not carry a plan that moves the object. The honest synthesis across the oracle ladder, the mitigation table, and this experiment is that contact planning faces *two* walls — an exploitable cost/optimizer interface (Secs. 5–6.2) and an insufficient frozen representation (this section) — and that *grounding is necessary but not sufficient*: neither hardening the cost, nor fine-tuning the encoder for grounding, nor removing the search adversary, crosses it alone.

**Scope and honest limits.** The exploitation mechanism is established on Metaworld, where simulator state lets us verify every claim against ground truth; on DROID we establish the diagnostic (scale-flat chance-floor grounding) and the predictor-axis repair, but closed-loop exploitation curves on real robots remain future work. The mitigation table covers the levers available on frozen pretrained checkpoints (readouts, metrics, adapters, LoRA fine-tuning, ensembles); full encoder retraining with planner-aware objectives is beyond our compute and remains open — our claim is about the *post-hoc* regime. The boundary-regime labels use an object-displacement proxy (no MuJoCo contact ground truth in the released data); mw-door-close is excluded as a proxy anomaly. The Reach parity check compares one released checkpoint over 16 episodes against a published 3-seed $\times$ 96-episode number — we read it as “reproduces, inside the CI”, not a seed-matched replication. Finally, an information-sufficiency question remains open beneath everything: vision-only latents may lack contact state outright (Zhang et al., 2026); our state-oracle row shows the *task* is solvable from privileged state, but whether any vision-only cost can be made pocket-free is not settled by this paper.

## 9 Conclusion

Contact-rich planning with JEPA world models does not fail for want of encoder scale, prediction accuracy, grounding, or readout precision — all were satisfied, measured, at some rung of our ladder while push stayed at 0–2/16. It fails because a test-time optimizer is pitted, hundreds of candidates per control step, against a learned cost whose residual-error pockets it is structurally guaranteed to find — and the same planner solves the same task 16/16 the moment the cost has no pockets. Repairs aimed at the average case (hardened readouts, fine-tuned encoders, ensembles) leave the worst case reachable; the training-side objective that repairs the predictor’s action channel is real and label-free but orthogonal to the cost-side wall. World models intended for planning should be evaluated the way their planners attack them: not by how well they

predict, but by how little they can be exploited.

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## A Regime stratification

Every per-regime table in the paper (Tables 1, 2, 7, 8) relies on assigning each transition ( $s_t \rightarrow s_{t+1}$ ) to exactly one of four regimes. The assignment is a *priority cascade*: the first matching rule wins, so the regimes are mutually exclusive by construction and the ordering encodes precedence (the gripper-actuation *event* outranks the contact *state* it may create). Thresholds are fixed constants (below), not fit to any outcome.

**Metaworld (simulator state).** The released HuggingFace dataset ships no MuJoCo contact flags but does ship the full 39-dim Metaworld-v2 observation, from which we recover the end-effector position (`state[0:3]`), gripper aperture (`state[3]`), and primary-object position (`state[4:7]`) exactly. For a transition:

1. **gripper actuation** if  $|\Delta_{\text{grip}}| > 0.10$  (the gripper is opening/closing — the grasp/release event itself);
2. else **contact manipulation** if  $\|\Delta_{\text{obj}}\| > 5 \text{ mm}$  (the object measurably moved — something pushed it);

3. else **pre-grasp** if  $\|ee - obj\| < 10$  cm (the hand is close to the object but has not yet moved it);
4. else **free space**.

“Contact” here therefore means *the object displaced*, an object-displacement *proxy* rather than a contact sensor; the caveat is carried on every Metaworld figure, and `mw-door-close` is excluded as a proxy anomaly (its hinge geometry violates the displacement assumption).

**DROID (real gripper sensor + latent-change proxy).** DROID-wrist has no object pose, so contact is read from the *real gripper sensor* (primary) and the approach cue from the latent-change proxy (secondary):

1. **gripper actuation** if  $|\Delta_{grip}| > 0.2$ ;
2. else **contact manipulation** if the gripper is held *closed* ( $grip_t > 0.5$  on the  $[0, 1]$  open→closed scale) — holding or contacting an object, a sensor signal that (deliberately) does *not* depend on visual motion, so a closed gripper resting on an object counts as contact even when the scene is momentarily static;
3. else (gripper open) **pre-grasp** if the scene is actively changing,  $\|\Delta z\| > 1.0 \times \text{median}\|\Delta z\|$  (an object entering/growing in the wrist view; the dataset-median baseline is the same one used for effect-conditioning);
4. else **free space** (gripper open, scene static).

Making contact gripper-primary rather than  $\|\Delta z\|$ -gated matters for the raw per-regime counts (the earlier motion-gated rule mislabeled  $\sim 54\%$  of closed-gripper *static-holding* steps as free space); it leaves the *headline* numbers unchanged, because those are effect-conditioned ( $\|\Delta z\| > \text{median}$ ) and the relabeled steps are the low- $\|\Delta z\|$  holding steps that effect-conditioning excludes either way.

**RoboCasa (grasp flags).** RoboCasa exposes an `object.in.hand` flag and proprioception, mapped to the same taxonomy: gripper actuation if  $|\Delta_{grip}| > 0.2$ ; else contact manipulation if `object.in.hand` at  $t$  or  $t+1$ ; else pre-grasp if  $\|ee - obj\| < 12$  cm (looser than Metaworld for the larger scene scale); else free space.

The three datasets thus share one taxonomy and one cascade ordering, differing only in the contact signal available (object displacement / gripper sensor / grasp flag). Source: `diagnosis/stratification/{metaworld,droid,robocasa}`

**Robustness to the threshold choice.** Because the pre-grasp locus (Table 1) is read off a fixed 5 mm/10 cm/0.10 cut, we re-ran classification and the hard-nearest-neighbour diagnostic (Sec. 3.1) under six perturbations — object-move  $\in \{2.5, 10\}$  mm, pre-grasp distance  $\in \{8, 12\}$  cm, gripper-delta  $\in \{0.05, 0.20\}$  — against the baseline. Pooled effect-conditioned CRA at pre-grasp moves by at most 0.037 (DINO-WM:  $[0.450, 0.488]$ ; JEPA-WM:  $[0.529, 0.558]$ ) across all seven configurations, and **pre-grasp remains the lowest-scoring regime, below both free space and contact manipulation, in every single configuration for both models** (`results/regime_robustness_summary.csv`, scripts/04+46). The locus of the collapse is a property of the transitions, not an artifact of the specific cut.

## B Boundary Blindness: definition

The instrument compares, on a neighbourhood of similar states, how much the *world’s* outcome fans out under nearby actions against how much the *model’s* prediction fans out under the same actions.

**Neighbourhood.** For each anchor transition  $i$  with latent  $z_i$  and action  $a_i$ ,  $\mathcal{N}(i)$  is its  $M=16$  nearest cached transitions in latent distance, radius-limited with relaxation (the same neighbourhood the hard-negative CRA sampler uses).  $o_j$  is the true scalar outcome of neighbour  $j$ : object displacement  $\|\Delta obj_j\|$  on Metaworld, the  $\|\Delta z_j\|$  proxy on DROID.

**Boundary regime.** A transition is *boundary* when small action changes produce large true-outcome changes across its neighbourhood:

$$\text{boundary\_score}_i = \frac{\text{std}_j o_j}{\text{mean}_j \|a_j - a_i\| + \varepsilon}, \quad (5)$$

thresholded at the 75th population percentile. Free-space drift has large  $\|\Delta z\|$  but smooth neighbourhoods and scores low.

**The two sensitivities and BB.** The world’s local action-sensitivity  $S_i^{\text{true}}$  is the dispersion of true outcomes across the neighbourhood; the model’s,  $S_i^{\text{model}}$ , applies each neighbour’s action to the *single* anchor state and measures the RMS spread of  $F(z_i, a_j)$  around its centroid. Each is z-scored over the population and

$$\text{BB}_i = \text{relu}(\tilde{S}_i^{\text{true}} - \tilde{S}_i^{\text{model}}), \quad (6)$$

pooled ( $n_b$ -weighted) over the boundary mask with trajectory-clustered bootstrap CIs. Because BB references the world, a high value cannot be an artefact of small one-step latent deltas.

## C The grounded-channel fix ladder (metric-level repair)

Deferred from Sec. 4: the model-side repair ladder that motivated the oracle ladder. **Nulls:** mixture-density predictor heads (soft-NLL/WTa/hard-EM/supervised,  $K \in \{2, 3\}$ ) beat their  $K=1$  NLL controls but do not move  $\text{BB}_{\text{boundary}}$  — the bifurcation spans 9.9  $L_2$  units against a 106 median residual (the geometry under-weights the boundary subspace  $10\times$ ), and the boundary’s action-dependence is unlearnable from expert-only data at the head level. Probe-subspace metric re-weighting redistributes BB without reducing it. **The probe chain:** object state is decodable from the frozen latent (V1 err 0.064 vs. sd 0.094), propagates through the predictor for *factual* actions (V2 0.059), but the *counterfactual* action→object response is noise (V3 spread-corr +0.035). **The repair:** a 0.5M grounded dynamics channel  $h(z_t, a) \rightarrow \widehat{\Delta\text{obj}}$  (mean/max-pooled tokens + action MLP), trained on cached latents with everything frozen, restores counterfactual tracking (0.035→0.682 DINO-WM; 0.702 JEPA-WM), halves boundary BB on both families (1.323→0.660; 1.280→0.620), and improves the closed-loop cost surface (Table 3) — while flipping zero successes, the observation Sec. 5 explains. Honest negative: the recipe does not transfer to DROID’s noisy whole-state proxy label at 2.1k transitions (rank-corr +0.708 but spread magnitude collapsed; BB unchanged) — the principle needs a learnable grounding label.

## D Probe, verifier, and mitigation-arm implementations

Every learned rung of the oracle ladder (Table 4) and mitigation table (Table 6) is small and trained on the cached latents with the encoder and predictor frozen. We give each arm’s spec so the “all repairs fail” claim can be audited.

**Object probes (the readout  $\rho$ ).** Two readouts decode object xyz from a single frozen latent, trained supervised on cached (latent, simulator object-position) pairs. The *pooled* probe is a mean+max-pooled-token MLP ( $2D \rightarrow 512 \rightarrow 256 \rightarrow 3$ , SiLU). The *spatial* probe prepends a learned CLS query to the patch tokens and routes it through a 2-layer, 6-head Transformer encoder before a linear head; being permutation-aware it can attend to *which* patch lit up, so it upper-bounds the spatial information the latent carries — it is the readout behind the 92%-within-5 cm decode in Table 4. The *expert* probe trains on expert-trajectory frames only; the *off-policy-hardened* probe mixes in latents of arbitrary candidate rollouts, which restores 91.5% within 5 cm on that off-policy distribution.

**On using  $\rho$  as the verifier.** The exploitation evidence (Table 5) decodes  $\rho$  on the CEM elites. Because  $\rho$  is itself OOD-sensitive — that sensitivity is precisely what is under test — we do not rest the claim on  $\rho$  alone: on the *executed* Metaworld episodes the simulator’s true object sits 13–30 cm from goal while the cost



reads near its minimum (Sec. 6.1), a ground-truth corroboration that does not pass through  $\rho$ . On real-robot data  $\rho$  (or a held-out verifier) is the only available instrument, which is why the evaluation protocol (Sec. 8) is stated for a generic verifier rather than privileged state. A tighter closed-loop version — decoding elites against simulator state directly, not  $\rho$  — is the natural next confirmation and is left as future work.

**Grounded object-dynamics channel  $h$  (Sec. 4, App. C).**  $h(z_t, a) \rightarrow \widehat{\Delta\text{obj}}$  is a mean+max-pooled-token MLP with an action-embedding branch ( $2D+64 \rightarrow 512 \rightarrow 256 \rightarrow 3$ ); the grounded cost integrates it along the rollout,  $\widehat{\text{obj}}_H = \rho(z_0) + \sum_t h(\hat{z}_t, a_t)$ , and scores  $\text{MSE}(z_H, z_g)/s_z^2 + \beta^2 \|\widehat{\text{obj}}_H - \rho(z_g)\|^2/s_g^2$ , the two terms rescaled by dataset scales  $s_z, s_g$  because they span  $\sim 98\text{k}$  vs. 3 dimensions (per-dim MSE would silently collapse the blend to object-only).

**Learned latent metric  $d_\theta$  (Tables 4, 6).** A goal-conditioned quasimetric over pooled tokens,  $d_\theta(z, z_g) = \|u_s(z) - u_s(z_g)\|_2 + \sum_i \text{relu}(u_a(z)_i - u_a(z_g)_i)$  — a symmetric part plus a directed residual (MRN-style), so  $d_\theta(z_g, z_g) = 0$  is the unique minimum. Trained *without* object labels on within-trajectory temporal order ( $d_\theta(z_t, z_{t'}) \approx t' - t$  for  $t < t'$ , cross-trajectory pairs pushed large), so the recipe is dataset-agnostic.

**Relearned representation adapter  $\varphi$  (Table 6).**  $\varphi = A(z)$  is a small head over the frozen patch tokens, trained with four losses so that plain  $L_2$  in  $\varphi$ -space is a harder-to-exploit cost: (i) *grounding* — the first obj-dim coordinates regress true object xyz; (ii) *counterfactual margin* — on hard-effect negatives (same  $z_t$ , different action, genuinely different true outcome) the two object estimates must be separated by at least their true gap, a margin constraint that targets residual-error *pockets* rather than average accuracy; (iii) *temporal ranking* confined to the non-object subspace; (iv) *adversarial* — previously-mined CEM elite frames are margin-pushed from their episode goal in the object subspace. This is the strongest post-hoc cost we built, and it still gates push 1/16.

**Encoder-LoRA (Table 6).** Zero-initialized LoRA adapters ( $B=0$ , so the injected model starts *exactly* at the frozen encoder) inserted into the encoder’s transformer blocks and trained with the  $\varphi$  losses; unlike  $\varphi$  — a function of the frozen  $z$ , so it can only read around entangled latents — this can separate them at the source. The adapters are toggleable, so one injected model serves both the frozen-baseline and LoRA arms; we train five seeds.

**Ensemble disagreement (Table 6).** Five independently fine-tuned encoder-LoRA seeds, with cost  $\text{mean}_k d_k^2 + \lambda \text{Var}_k[d_k]$  — the standard pessimism penalty — swept over  $\lambda \in \{0, \frac{1}{2}, 1, 2\}$ .

## E Reproduction integrity

Closed-loop sim evaluation is fragile, and getting the baseline wrong would invalidate the parity argument. Three environment-side bugs, found and fixed, all model-agnostic: **(1)** Metaworld’s offscreen renderer is constructed in the env constructor, so camera/size assignments afterwards were silently ignored (default camera at 480 px vs. the trained **corner2** at 224 px; symptom: one-step prediction error  $8.5\times$  dataset level). We re-create the renderer explicitly, mirroring the upstream wrapper. **(2)** The released training frames are vertically flipped relative to today’s **corner2** render; pixel-calibration against dataset initial frames scores the flipped wrapper-camera render at MSE 71.6 vs.  $\geq 3000$  for every unflipped candidate. **(3)** The success flag fires at a 5 cm radius but the scripted expert refines to  $\sim 1$  cm; taking the goal frame at first-success rather than the expert’s final frame parked the planner just outside the radius. **Verification:** after the fixes, one-step prediction error is  $1.4\times$  dataset level, latent NN ratio 0.97, and replaying dataset actions reproduces the proprioceptive trajectory to 1.5 mm median end-effector error — only the visuals had drifted, never the physics. The reach baseline then recovers from near-zero to 37.5% (Table 3), inside the published CI.

## F AUG and ECS: the raw-gap corroboration

CRA leads the main text because, as a ranking, it is cross-model comparable. The two raw-gap metrics corroborate it: AUG (prediction-error gap between permuted and factual actions;  $\approx 0$  iff actions ignored) and ECS (AUG on effect-bearing transitions), both on the hard nearest-neighbour family.

Table 8: AUG / ECS, hard nearest-neighbour family, per regime ( $n_{\text{eff}}$ -weighted pools; dash = no effect-bearing transitions). Raw latent gaps are *not* comparable across models (latent scales differ); the cross-model statement lives in CRA (Tables 1, 2).

| model                                                   | free space   | pre-grasp    | gripper      | contact       |
|---------------------------------------------------------|--------------|--------------|--------------|---------------|
| <i>DROID</i> (real robot; AUG / ECS)                    |              |              |              |               |
| DINO-WM (22M)                                           | 0.038/–      | 0.011/0.0105 | 0.038/0.0645 | 0.070/0.0094  |
| JEPA-WM (300M)                                          | 0.000/–      | 0.000/0.0002 | 0.001/0.0004 | 0.001/–0.0011 |
| V-JEPA-2-AC (1B)                                        | 0.006/–      | 0.002/0.0016 | 0.003/0.0026 | 0.003/–0.0059 |
| V-JEPA-2-AC-OSS (1B)                                    | 0.004/–      | 0.001/0.0010 | 0.001/0.0005 | 0.003/–0.0030 |
| <i>Metaworld</i> (sim; AUG / ECS; pooled over 12 tasks) |              |              |              |               |
| DINO-WM (22M)                                           | 0.088/0.1007 | 0.051/0.0997 | –/–          | 0.081/0.0893  |
| JEPA-WM (300M)                                          | 0.089/0.1020 | 0.058/0.1114 | –/–          | 0.095/0.1052  |

The effect-conditioned action gap is statistically indistinguishable from zero on the real-robot data across the full  $45\times$  scale range — the chance-level CRA reflects genuine action-ignoring, not a ranking quirk. Source: `results/{droid,metaworld}.diagnostic.csv`, `hard_nn` rows.

## G Full per-task tables

Per-task Boundary-Blindness dumps, per-arm mitigation episode tables (phase C/D/F/G logs), the overoptimization-sweep curve tables, and the Phase-H seed-sweep confidence intervals are provided in machine-readable form under `diagnosis/results/` (see `diagnosis/docs/CLAIMS_EVIDENCE.md` for the row-to-table mapping).